



Vivekanand Education Society's Institute of Technology

(Autonomous Institute Affiliated to University of Mumbai, Approved by AICTE & Recognised by Govt. of Maharashtra)
NAAC accredited with 'A' grade

Semester: VI

Title of the Project:
Sound Level Detector Using Electret Microphone Sensor

Subject : Sensor Lab

Group Members:

Member 1 : 03 Kartik Bhat

Member 2: 13 Spandan Deb

Teacher Name: Mr. Abhishek Chaudhari



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Introduction to Project

- Accurate sound level detection is essential for applications ranging from noise pollution monitoring to interactive installations and educational tools.
- This project utilizes an electret microphone sensor, a compact and cost-effective component, to capture and convert sound waves into electrical signals.
- By integrating the electret microphone with an Arduino microcontroller, the system processes analog sound signals and controls LEDs to visually indicate varying sound intensities.
- The project aims to develop a reliable and efficient sound level detection system, leveraging the microphone's precision and real-time response capabilities.



Objectives of the project

- **Accurate Sound Sensing:** Implement the electret microphone sensor to measure sound levels precisely, even in environments with varying noise levels.
- **Real-Time Sound Detection:** Design a system capable of detecting and responding to sound intensity changes in real-time, providing immediate feedback through visual indicators.
- **Easy Microcontroller Integration:** Interface the electret microphone with platforms like Arduino, utilizing simple analog input for seamless and flexible operation.
- **Compact and Efficient Design:** Develop a lightweight and energy-efficient solution suitable for portable or embedded systems, ensuring minimal power consumption.
- **Customizable Threshold Settings:** Implement adjustable threshold settings to allow users to define specific sound levels for triggering different responses, enhancing the system's adaptability to various environments and user preferences.



Requirements of the system (Hardware, software)

HARDWARE REQUIREMENTS

- **Electret Microphone:** Condenser microphone with 20 Hz to 20 kHz frequency response, -44 dBV/Pa sensitivity, 2.2 k Ω impedance, operating at 3V to 10V DC.
- **Arduino Uno (Microcontroller):** ATmega328P-based, 5V operating voltage, 16 MHz clock speed, 32 KB flash memory, 2 KB SRAM, 1 KB EEPROM, 14 digital I/O pins (6 PWM), 6 analog input pins.
- **Jumper Wires:** Male-to-male, male-to-female, or female-to-female types, typically copper with insulation, varying lengths for circuit connections.
- **Breadboard:** Solderless prototyping board, sizes like 400 or 830 tie-points, plastic with conductive metal strips for temporary circuit assembly.
- **LED Bulbs:** Light Emitting Diodes in various colors, typically 1.8V to 3.3V forward voltage, 20 mA forward current, varying luminous intensity.
- **Resistor:** Commonly 220 Ω for LEDs, $\pm 5\%$ or $\pm 1\%$ tolerance, 1/4 watt or 1/2 watt power rating, carbon or metal film material.
- **Capacitor:** Ceramic or electrolytic types, commonly 0.1 μ F capacitance, voltage ratings from 16V to 50V, $\pm 5\%$ or $\pm 10\%$ tolerance for signal smoothing.



Requirements of the system (Hardware, software)

SOFTWARE REQUIREMENTS

- **Arduino IDE** – Used to write, compile, and upload code to the Arduino board



Literature Survey

Sr.No	Title	Author	Publish Date	Description
1]	Smart Embedded Framework using Arduino and IoT for Real-Time Noise and Air Pollution Monitoring and Alert system	Janeera.D.A, Poovizhi.H, Sheik Haseena.S.S, Nivetha.S	April 2021	• The system integrates sensors for sound, gas, humidity, and temperature, connected to an Arduino microcontroller, with data transmitted via Wi-Fi to the ThingSpeak cloud platform for remote access and analysis. • Testing in various pollution environments demonstrated effective monitoring and alerting capabilities, making it suitable for industrial areas, residential zones, hospitals, and regions with heavy traffic, while being cost-effective, efficient, and user-friendly.
2)	An IoT Solution for Secured and Remote Sound Level Monitoring	Milica Lekić, Jovan Galić, Sonja Matić	March 2019	▪ The system employs a Raspberry Pi as the MQTT broker, facilitating data transfer between the publisher (Arduino with sound sensor) and subscriber (PC). The implementation ensures secure and efficient data communication, with the MQTT protocol enabling real-time monitoring and alerts.



Literature Survey

Sr.No	Title	Author	Publish Date	Description
3]	The Working Principle Of An Arduino.	Yusuf Abdullahi Badamasi	December 2014	• The hardware components of the Arduino board, such as the USB plug, power supply, microcontroller, analog and digital pins, and reset button, are discussed in detail. The paper also covers the Arduino IDE (Integrated Development Environment), including its command area, text area for coding, and message window for feedback. • Practical examples, such as a sensor alarm system and a capacitance measuring circuit, are provided to illustrate how Arduino can be used in real-world projects. • The paper concludes by summarizing the key aspects of Arduino's hardware and software, making it a comprehensive guide for beginners

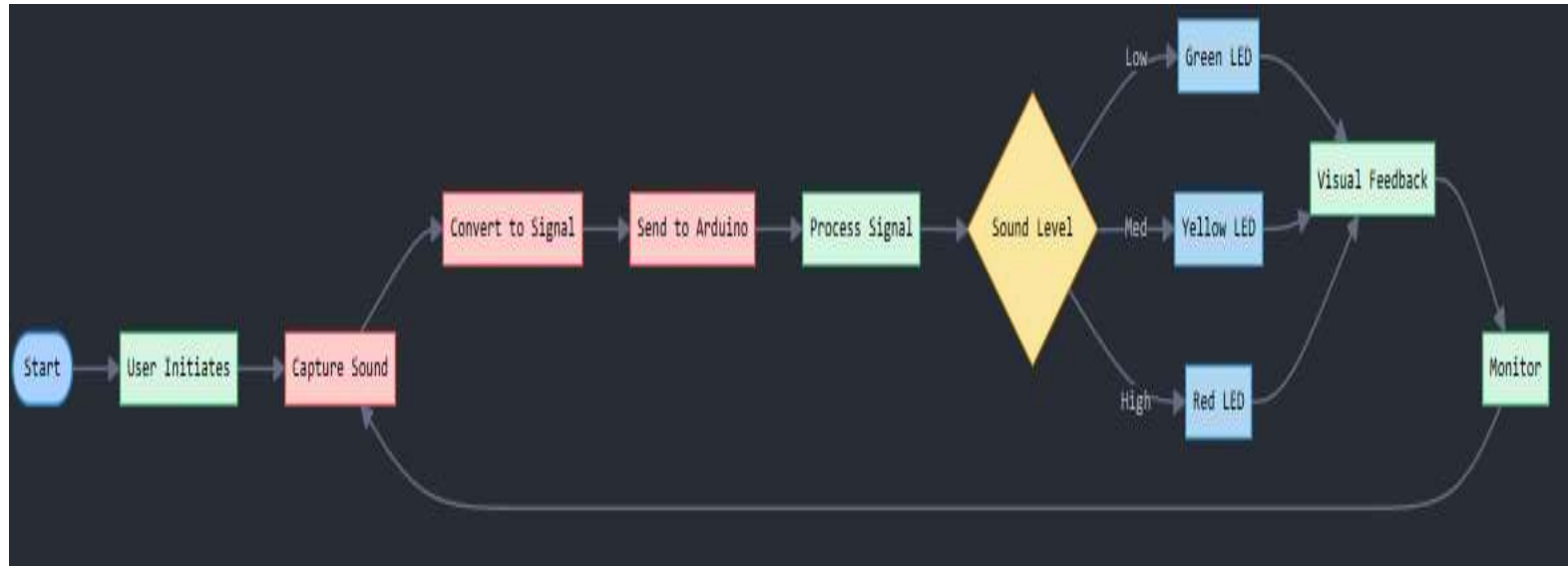


Proposed System

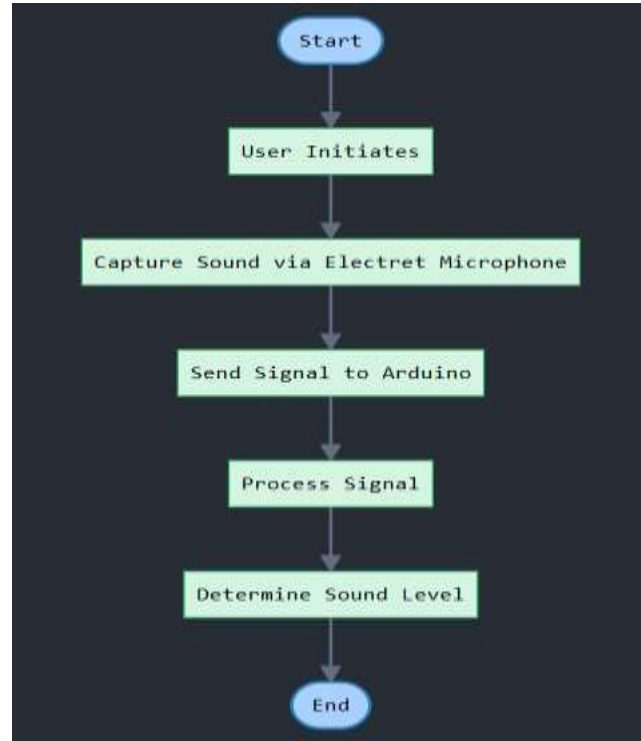
The proposed system for the Sound Level Detector project is designed to capture and respond to varying sound levels using an Arduino microcontroller and an electret microphone.

- The electret microphone captures sound waves and converts them into electrical signals.
- The Arduino reads these signals through its analog input pin and processes them.
- Based on predefined thresholds, the Arduino controls the LEDs to indicate the detected sound level.
- The system can be calibrated using the Serial Monitor to adjust thresholds for optimal performance in different environments.

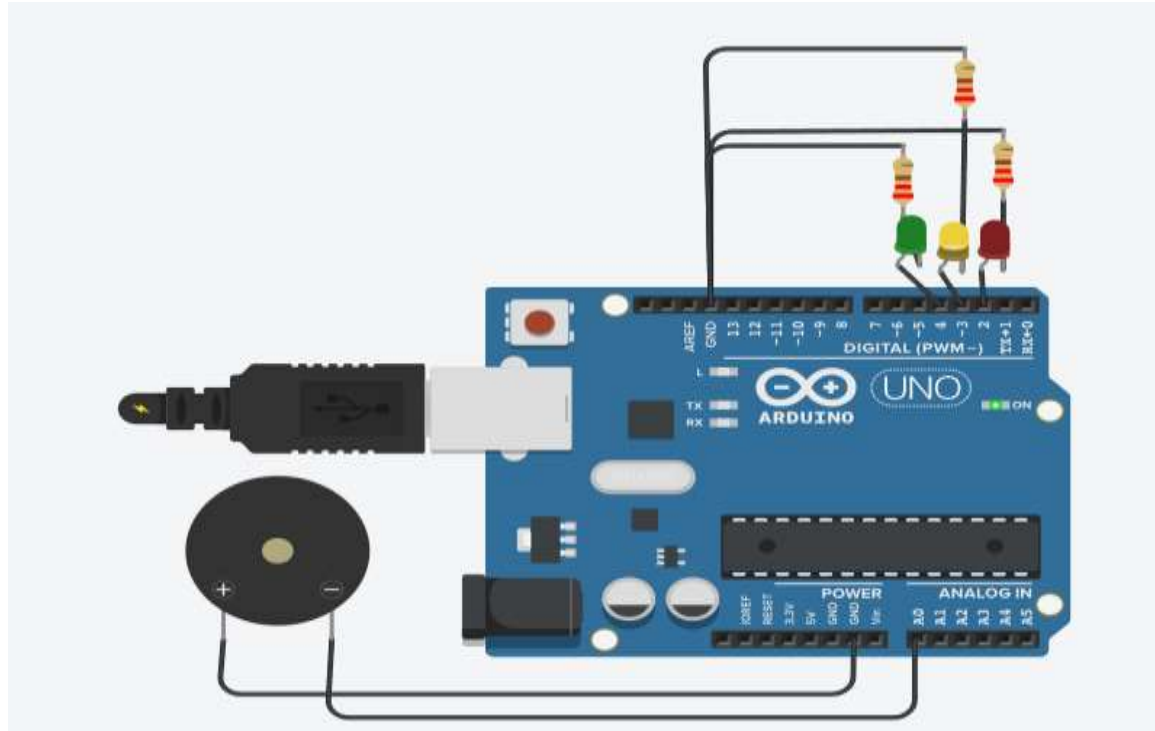
Proposed Design



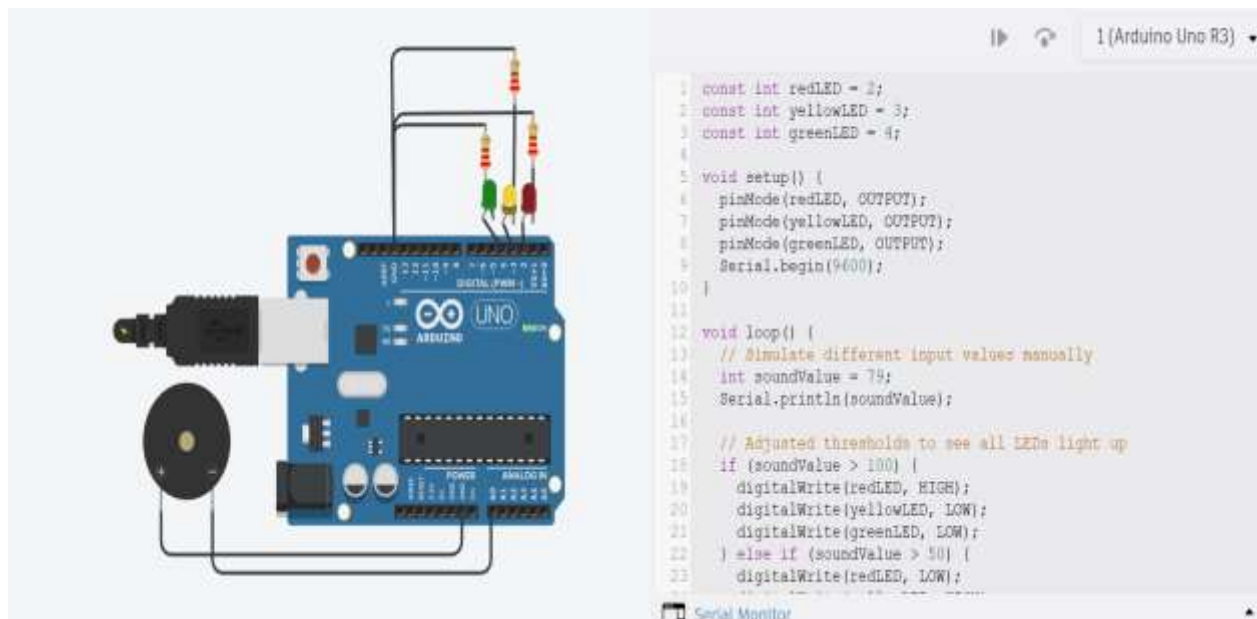
Proposed Design [Flowchart]



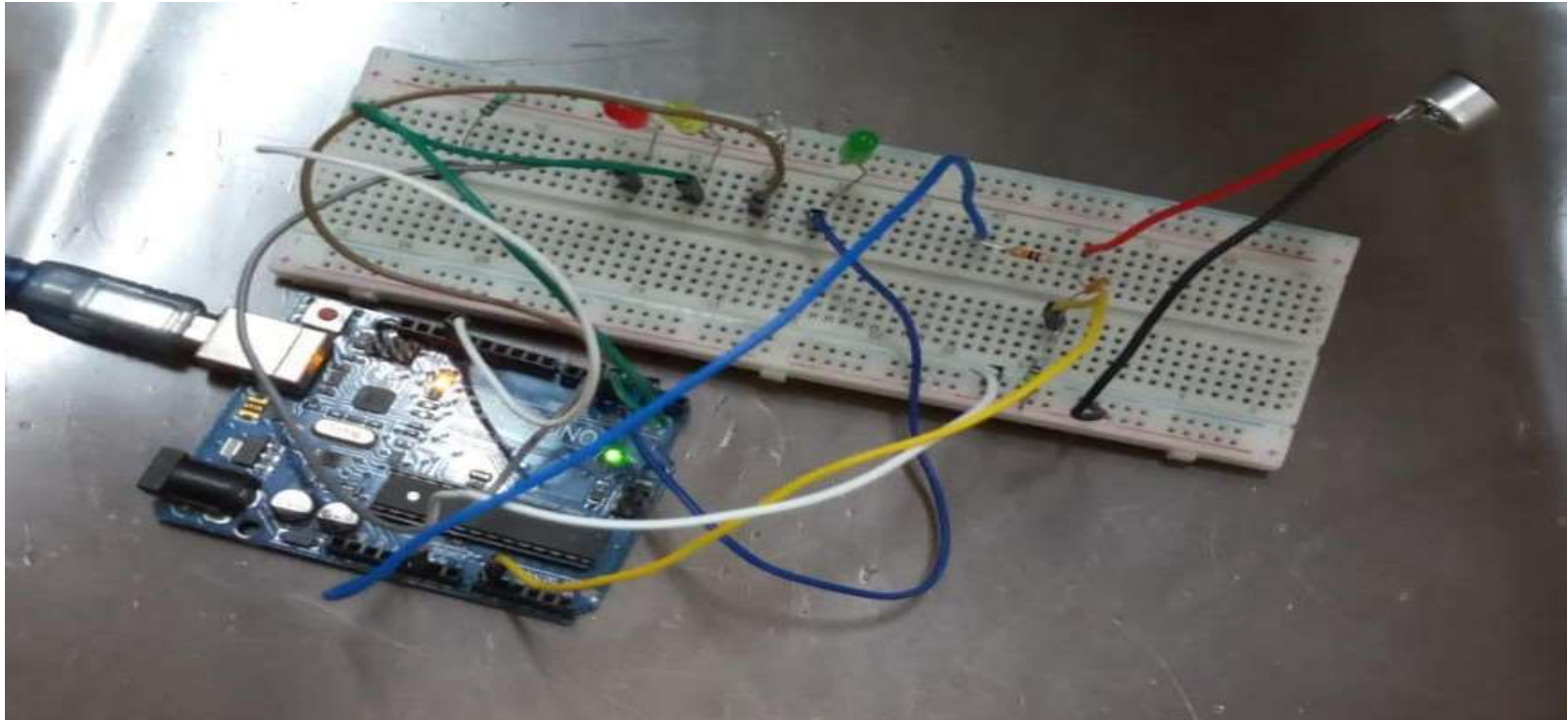
Simulation



Simulation



Implementation





Result Analysis

- The electret microphone accurately captured varying sound levels and consistently generated analog signals across different environments.
- The Arduino processed sound signals efficiently and triggered LEDs in real-time based on predefined thresholds.
- LED indicators responded clearly and immediately, providing reliable visual feedback aligned with sound intensity.
- The overall setup was stable and reliable across multiple tests, making it ideal for educational and DIY applications.



Conclusion

- Successfully developed a functional and interactive sound level detection system.
- Integrated an electret microphone with Arduino for real-time monitoring and data collection.
- Enabled accurate detection of varying sound intensities in different environments.
- Showcased potential applications in noise pollution monitoring, smart environments, and educational demonstrations.
- System is low-cost, easy to replicate, and ideal for DIY electronics and IoT projects.
- Laid a strong foundation for future enhancements such as data logging, wireless transmission, and machine learning integration.



References

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- [4] Zhao, Y., Deng, G., Zhang, L., Di, N., Jiang, X., & Li, Z. (2021). Based investigate of beehive sound to detect air pollutants by machine learning. Ecological Informatics, 101246.